

Muhammad Ahmad
FA23-BCE-113

Lab-Task

```
n = 9;
A = zeros(n,n)

for i = 1:n
    A(i, i) = 4 % Main diagonal

    % lower diagonals
    if i > 1
        A(i, i-1) = 1
    end
    if i > 3
        A(i, i-3) = 1
    end
    if i > 5
        A(i, i-5) = 1
    end
    if i > 7
        A(i, i-7) = 1
    end

    % upper diagonal
    if i < n
        A(i, i+1) = 1
    end

    if i <= n - 3
        A(i, i+3) = 1
    end

    if i <= n - 5
        A(i, i+5) = 1
    end

    if i <= n - 7
        A(i, i+7) = 1
    end
end
```

```
A =

     4     1     0     1     0     1     0     1     0
     1     4     1     0     1     0     1     0     1
     0     1     4     1     0     1     0     1     0
     1     0     1     4     1     0     1     0     1
     0     1     0     1     4     1     0     1     0
     1     0     1     0     1     4     1     0     1
     0     1     0     1     0     1     4     1     0
     1     0     1     0     1     0     1     4     1
     0     1     0     1     0     1     0     1     4
```